

Stanley College of Engineering and Technology for Women(A)

B.E I Semester (Backlog) Examinations May-June-2023

Mathematics-I
(Common to all branches)

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- | | | | |
|---|---|-----|-----|
| 1 | Write the statement of Cauchy's nth root test | 2 M | L 1 |
| 2 | Verify Rolle's theorem for $f(x) = x^3$ in $[1,3]$ | 2 M | L 1 |
| 3 | Find first and second order partial derivatives of $x^3 + y^3 - 3axy$ | 2 M | L 1 |
| 4 | Find the Curl f if $f = xyi + yzj + zxk$ | 2 M | L 1 |
| 5 | Find the first two iterations for $x^3 - 2x - 5 = 0$ by the method of Bisection correct to three decimal places | 2 M | L 2 |

PART-B

Note: Answer all questions

5 X 10= 50M

- | | | | |
|---|--|------|----|
| 6 | a. Test the convergence of the series $\sqrt[n]{n+1} - \sqrt[n]{n-1}$ | 5 M | L3 |
| | b. Test the convergence of the series $\sum (nx/(n+1))^n$ | 5 M | L3 |
| | OR | | |
| | c. Test the convergence of the series $\sum (1.2.3.....n)/(3.5.7.....(2n+1))$ | 5 M | L3 |
| | d. Examine the convergence of the series $1 - 1/\sqrt{2} + 1/\sqrt{3} - 1/\sqrt{4} + \dots$ | 5 M | L3 |
| 7 | a. Prove that $\pi/6 + 1/(5\sqrt{3}) < (\sin)^{-1}(3/5) < \pi/6 + 1/8$ | 5 M | L4 |
| | b. Verify Cauchy's Mean value theorem for $f(x) = \sin x$ and $g(x) = \cos x$ on $(0, \pi/4)$ | 5 M | L3 |
| | OR | | |
| | c. Show that the radius of curvature at any point of the cycloid $x = (a + \sin\theta)$ and $y = a(1 - \cos\theta)$ is $4a \cos(\theta/2)$ | 5 M | L4 |
| | d. Find the evloep of a system of concentric and coaxial ellipses of constant area. | 5 M | L4 |
| 8 | a. Find the maximum and minimum distances of the point $(3,4,12)$ from the sphere $x^2 + y^2 + z^2 = 4$ | 10 M | L5 |

OR
1

P.T.O

- c. If $u = F(x-y, y-z, z-x)$ then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$ 5M L3
- d. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ then show that $(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z})^2 u = (-9)/(x+y+z)^2$ 5M L4
- 9 a. If $F = (x+y+1)i + j - (x+y)k$, show that $\text{Curl } F = 0$ 5M L3
- b. Show that the vector $F = 3y^4z^2i + 4x^3z^2j + 3x^2y^2k$ is solenoidal. 5M L3
- OR
- c. Verify Green's theorem for $\int_C (xy + y^2)dx + x^2dy$ where C is bounded by $y = x$ and $y = x^2$ 10M L4
- 10 a. Apply Gauss elimination method to solve the equations $x + 4y - z = -5$, $x + y - 6z = -12$, $3x - y - z = 4$ 5M L3
- b. Find the root of the equation $\cos x = x e^x$ using the regula-falsi method correct to three decimal places. 5M L3
- OR
- c. Find the positive root of $x^4 - x = 10$ correct to three decimal places, using Newton-Raphson method. 5M L3
- d. Find the interpolating polynomial for (0,2), (1,3), (2,12) and (5,147) using Lagrange's interpolating formula. 5M L2

Hall Ticket No-

Code: S123514

Stanley College of Engineering and Technology for Women(A)

B.E(IT) I Semester (Backlog) Examinations May-June-2023

Basic Electrical and Electronics Circuits

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

1 Define Ohm's Laws.

5 X 2=10M

2M BTL1

2 Define R.M.S. value of A.C.

2M BTL1

3 List the Applications of PN junction diode.

2M BTL1

4 Compare CB,CE,CC in any four parameters.

2M BTL5

5 Write ideal characteristics of Ideal Op-Amp.

2M BTL4

PART-B

Note: Answer all questions

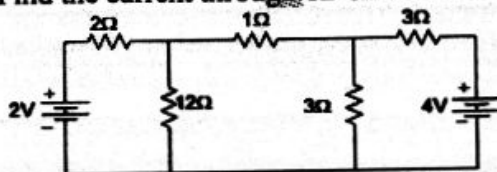
6 a. State and explain Superposition theorem. What are the limitations of Superposition theorem?

5 X 10= 50M

5M BTL2

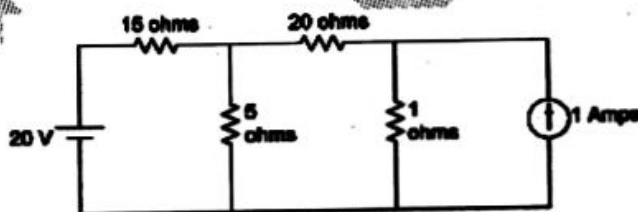
b. Find the current through 12-ohm resistor for the given circuit.

5M BTL3



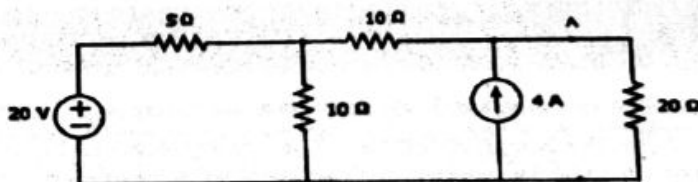
c. Solve for current in 5 ohms resistor by principle of super position theorem shown in circuit below

5M BTL3

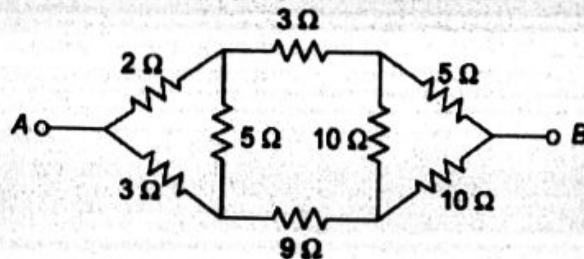


d. Find the current through 20 ohms resistor using Norton's theorem

5M BTL3



- 7 a. Show that the line current is equal to $\sqrt{3}$ times the phase current in delta connected system? 5M BTL2
- b. Find the voltage to be applied across AB in order to drive a current of 5A into the circuit by using star-delta transformation. 5M BTL3



OR

- c. Derive the expression for average and RMS Values of a Sinusoidal waveform. 5M BTL1
- d. Derive the expression for current, phase angle, power factor and power for R-L-C series circuit. 5M BTL6
- 8 a. Draw the circuit diagram of PN junction diode in forward and reverse bias. Draw the V-I. 5M BTL1
- b. Draw and Explain the Block diagram of Full wave rectifier circuit with suitable wave forms. 5M BTL4
- OR
- c. Explain the working of half wave Rectifier with a neat circuit diagram along with necessary graphs. 5M BTL2
- d. Differentiate between Avalanche and Zener breakdown. 5M BTL2
- 9 a. Explain the working of JFET and also describe the drain and transfer characteristics. 5M BTL2
- b. Write the difference between BJT and JFET? 5M BTL5
- OR
- c. Explain the construction of NPN transistor and its operations? 5M BTL2
- d. Derive the relationship between α , β and γ 5M BTL3

- 10 a. Draw the circuit diagram of RC- phase shift Oscillator and explain its working. Write the equation for frequency of oscillations. 5M BTL4
- b. Explain the integrator and differentiator using op-amp. Derive its required equations. 5M BTL2

OR

- c. What is piezoelectric effect? Explain the operation of crystal oscillator. 5M BTL1
- d. Derive the properties of negative feedback amplifier. 5M BTL6
